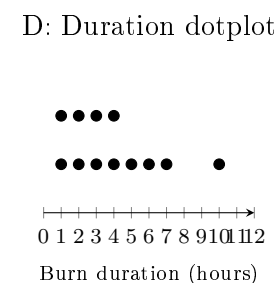
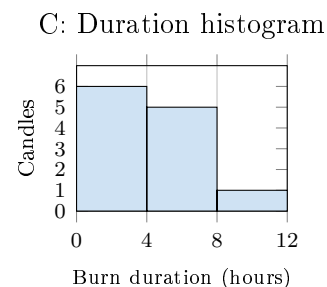
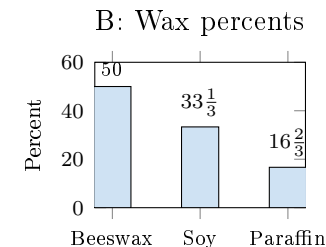
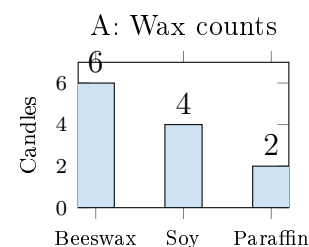


Candle Tests: Choosing the Evidence

Unit 1 · Topics 1.4–1.8 (CED 1.4–1.8) · THE PROBLEM

Suppose a candle workshop tests 12 candles. It records wax type and burn duration in hours. A and B summarize wax type for these same 12 candles; C and D summarize their burn durations. The links between wax type and duration are not shown. Histogram bins include the left endpoint and exclude the right endpoint; each dot represents one candle.

The report must identify the most common wax type and the number of candles that burned exactly 5 hours. An editor proposes: *Keep only B and C. B changes which wax is most common because its bars are taller than A's. C shows five candles at exactly 5 hours, so D adds nothing needed.*



FIRST TAKE — one sentence before you work the parts

Which part of the proposed report needs the closest check, and why?

- DOK 1** (a) Name the categorical variable in A and give one of its categories. Name the quantitative variable in C and its unit.
- DOK 2** (b) Convert each count in A to a percent and compare with B. Explain whether these graphs represent different category distributions. **Check your response:** ☐ all three categories; ☐ total used; ☐ axis units.
- DOK 2** (c) Compare what the bar from 4 to 8 in C tells you with the dots over that interval in D. State the interval count and every individual duration there, including repeats. Explain the difference in precision, using candles and hours.
- DOK 3** (d) ★ Evaluate the editor's plan in a short paragraph. Defend a set of displays sufficient for both reporting questions and correct any unsupported claims using numerical evidence. The teacher scores this response E/P/I by hand; nothing is automatically scored.

Self-paced bonus problem. Work (a)–(d) from this sheet; turn it in whenever you finish.